

Kaitlyn Forks

Full Stack Software Engineer in St. Louis, MO

A published peer-reviewed Biologist turned Software Engineer, Kaitlyn learned R and Python while researching reptiles and amphibians, using code to analyze ecological data and build predictive models. These foundational skills led her to full-stack development, where she now specializes in lightweight architecture, relational database design, and ensuring scalable applications using Node.js and Angular.

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EDUCATION

LaunchCode, Software Development Bootcamp — January 2020

University of Missouri St. Louis, MA, Science Education — May 2019

Texas A&M University, BA, Ecology & Evolutionary Biology — May 2017

WORK EXPERIENCE

Software Engineer — Appriss Insights / Equifax

Oct 2022 - December 2024

- **Led the Node.js / Angular front-end development for MOVAN, an internal ETL (Extract, Transform, Load) application** that automates real-time notifications to keep victims informed about changes in perpetrator status.
- **Collaborated with backend engineers to optimize data ingestion and transformation workflows**, improving processing time and system efficiency.
- **Developed front-end components that optimized large-scale data ingestion and transformation**, allowing users to manage high-volume data jobs intuitively.
- **Enhanced data visualization to improve clarity for internal users**, reducing errors in data transformation.
- **Automated form validation and user input verification**, increasing data integrity and reducing manual errors.

Full Stack Software Engineer — Anchor Yacht Charters

May 2021 - Oct 2022

- **Architected the Boat Owner Calendar and Charter Booking features**, allowing full customization of schedules, eliminating double bookings, and removing employees from the scheduling workflow.
- **Automated payout systems using Stripe**, streamlining financial workflows, reducing payment processing time, and ensuring secure financial data handling.
- **Designed a microservice for tax and fee calculations**, improving system performance and ensuring compliance with local regulations.
- **Integrated Google Maps API to enhance real-time location services**, increasing bookings and customer satisfaction.
- **Worked directly with CEOs and stakeholders** to deliver features that contributed to positive year-over-year growth in charter bookings.

SKILLS

PROGRAMMING LANGUAGES

- Javascript
- Typescript
- Python
- HTML/CSS
- Java
- R

FRAMEWORKS

- Node.js
- Angular
- NestJS
- Flask
- SQLAlchemy
- Sequelize ORM

DATABASES

- MySQL
- SQLite
- PostgreSQL

Third Party Integrations/ APIs

- Stripe
- Google Maps API
- AWS Cloud Services
- Hubspot & Twilio

Workflows, Methodologies, and More

- Software Consulting
- Database Design
- End to end project ownership
- Test Driven Development
- Agile Workflow

WORK EXPERIENCE CONTINUED

Junior Software Engineer — Varsity Tutors LLC / Nerdy NYSE

Dec 2020 - May 2021

- **Spearheaded the automation of manual tasks during my role on the business side**, reducing my daily workload by 4 hours through developer tools automation, which led to my transition to the engineering team.
- **Automated the creation of scalable landing pages in YAML by linking a CSV input process**, enabling business teams to generate large batches of B2B marketing pages, improving campaign turnaround times.
- **Integrated Dato CMS to enable non-technical staff to manage landing pages independently**, freeing up the engineering team to focus on SEO optimization and custom reusable components.
- **Focused on improving lead generation and conversion by building and testing reusable front-end components in Ruby on Rails**, contributing to increased engagement across marketing campaigns.

ADDITIONAL PROJECTS

Lead Full Stack Engineer and Tech Consultant — Spite House Studio Party Games

2022, 6 month freelance project

- **Participated in a Kickstarter campaign that raised over a quarter of a million dollars** to create and distribute a card game based on principles of improvised comedy.
- **Developed the entire e-commerce platform alongside one other engineer, using NestJS, Angular, and SQL** to support secure payments, order fulfillment, and product browsing.
- **Integrated Stripe for payment processing and HubSpot** for automating business email workflows, streamlining customer communications and marketing efforts.
- **Connected distribution and shipping APIs from ShipStation and "Fun Again" Board Game makers to automate order fulfillment and customer service interactions**, reducing manual processing time and ensuring timely deliveries.
- **Built dynamic product carousels, shopping cart logic, and coupon code functionalities**, enhancing the user experience and driving increased sales conversions.
- **Collaborated closely with the CEO and product team to ensure the e-commerce platform aligned with business goals**, resulting in a successful launch and increased revenue.

RESEARCH & ACADEMIC EXPERIENCE

Research Assistant — Ecological Systems Laboratory, Texas A&M University

June 2015 - May 2017

- **Developed population models using STELLA and R** to analyze ecological data and predict population trends.
- **Presented research** on trapping methods and fire ecology at national symposia.

Researcher/Volunteer — Genetics Laboratory, Texas State University

June 2015 - May 2017

- **Used R to analyze** amphibian and reptile capture data, applying statistical models to track species movement and health indicators.
- **Conducted data collection and genetic sampling** in collaboration with faculty on conservation projects.

Intern — United States Geological Survey, Gainesville, Florida

July 2016 - August 2016

- **Used R and Python for hierarchical data analysis**, developing code to predict home ranges of tree frogs.
- **Assisted in capture and genetic sampling** of amphibians for research on disease transmission.

PEER REVIEWED PUBLICATIONS & RESEARCH AWARDS

Prevalence of *Batrochochytrium dendrobatidis* in Two Sympatric Tree Frog Species, *Hyla cinerea* and *Hyla versicolor*

Herpetological Review, Dec 1, 2016

A Simple Conservation Tool to Aid Restoration of Amphibians Following High-Severity Wildfires: Use of PVC Pipes by Green Tree Frogs (*Hyla cinerea*) in Central Texas, USA

Diversity 13, no. 12: 649 · Dec 6, 2021

Applied Biodiversity Conservation Scholar — National Fish and Wildlife Foundation (NFWF)
2016

2nd Place Poster Presentation — Ecological Integration Symposium
April 2016

2nd Place Poster Presentation — Student Research Week, Texas A&M University
April 2016

Undergraduate Research Scholar — Department of Wildlife and Fisheries, Texas A&M University
August 2015 - May 2016

Undergraduate Research Grant Recipient — Texas A&M University
August 2015 - December 2015

EXTRAS

Improv Comedy: Regular performer at The Improv Shop in St. Louis, where I've been on a house team since 2018. I've also performed at the Flyover Comedy Festival (4x) and traveled out of state to perform at festivals like the Omaha Comedy Festival.

Saint Louis Zoo: Spent nights sleeping in the sea lion tunnel and developed a curriculum to help Girl Scouts earn their badges. Worked in the Department of Audience Research and did Participatory Asset Mapping to prepare for the 2027 opening of WildCare Park, a world-class safari park.

Creepy Crawly Enthusiast: My dad co-wrote the Field Guide to Texas Snakes, and growing up in a house with a snake farm inspired my lifelong love for reptiles and amphibians.

Avid Hobby-ist: When I'm not coding, I'm tending to my garden, playing the ukulele, baking bread or doing some other culinary art!

Skydiving on My First Flight: The first time I flew on a plane, I jumped out of it—skydiving to kick off my flying adventures.